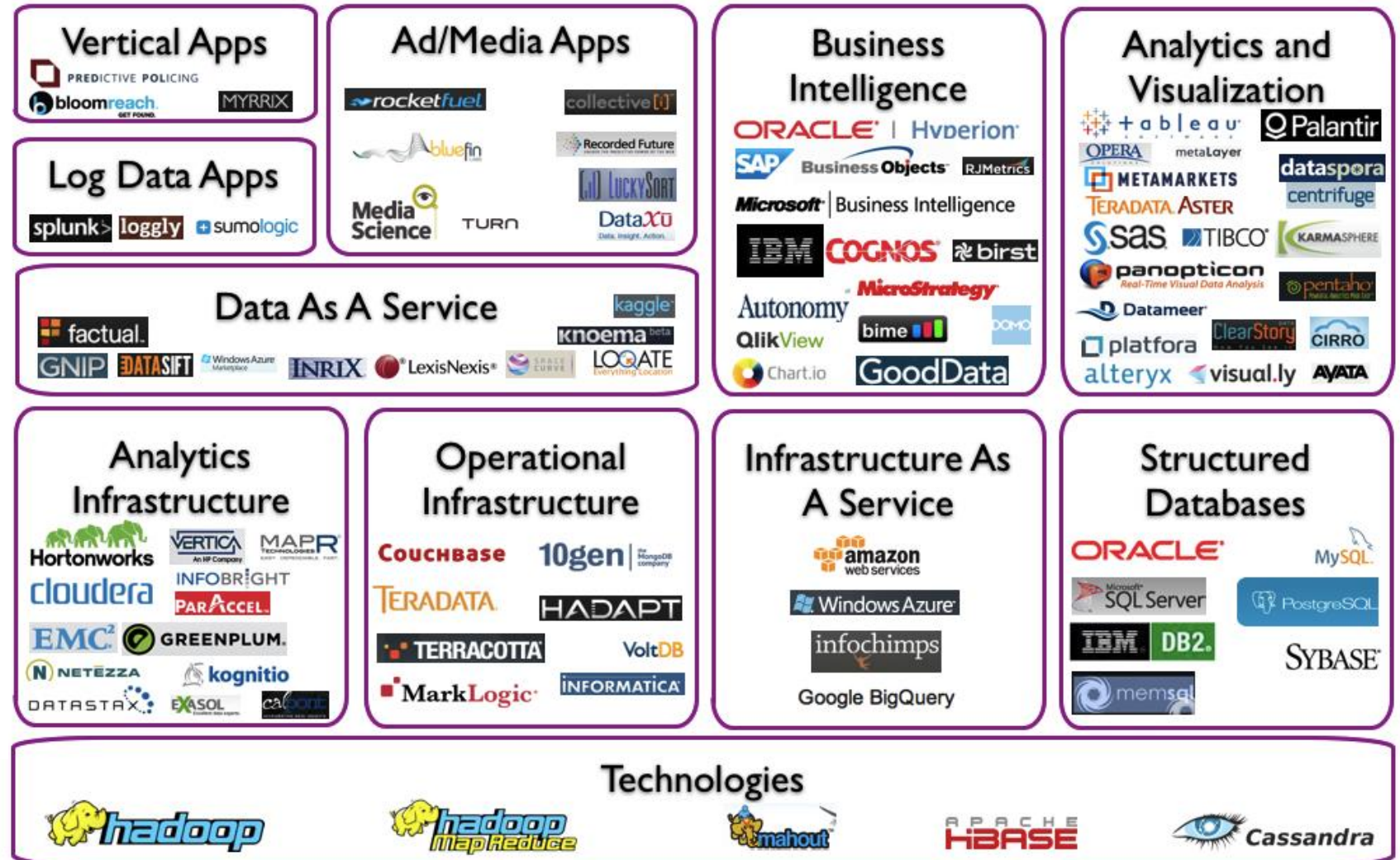
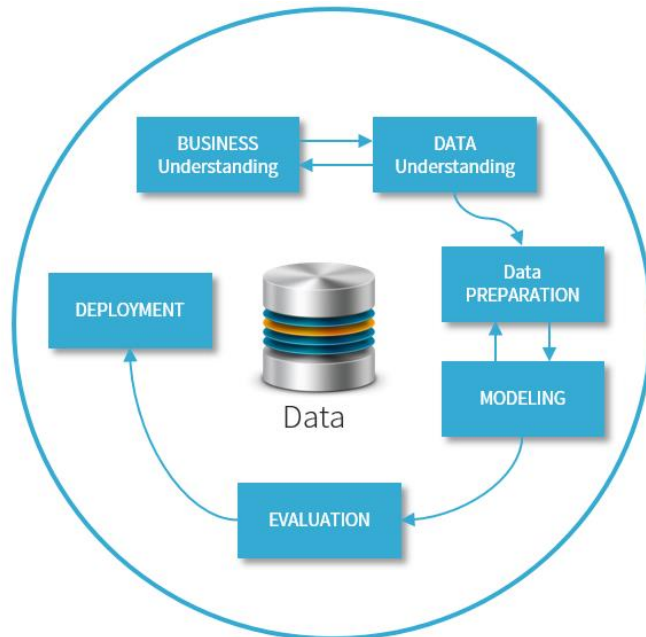




Introduction to Database Systems

2024-Fall

Big Data Ecosystem



Resources



CMU 15-445/645



<https://db.cs.cmu.edu>

CS w186 Introduction to Database Systems

Prof. Joe Hellerstein



Textbook:

- Nengbin Wang, "Textbook of Database Systems"
- Database Management Systems,
- Database System Concepts (7th Edition)
- Database Systems: The Complete Book



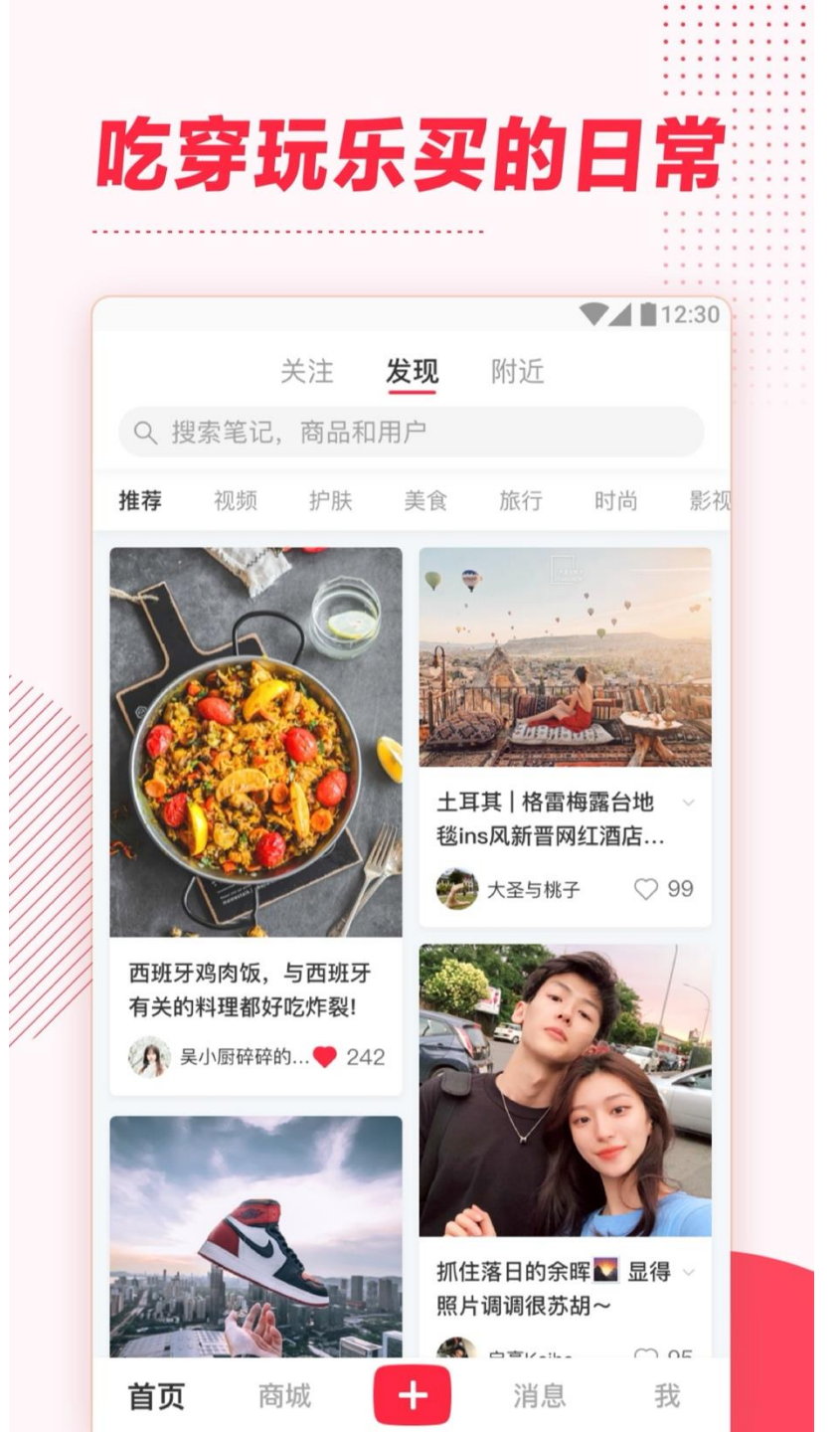
Essential Queries

- **Why** take this class?
- **What** is this class all about?
- **Who** is running this?
- **How** will this class work?



Why-R1: Utility

- This class is very, very useful
 - Data processing backs essentially every app
 - Databases of one form or another back most apps
 - The ***principles*** taught in this class back nearly everything in computing





Why-R1: Utility

- The **fundamentals** of this class are (and will remain) central to participating in this new and more data-centric world
 - Many of the details and technologies will change in the coming years
 - Be prepared to generalize from what you learn here
 - Keep learning new things
- This material will empower you.



Why-R2: Centrality

- Data is at the center of modern society
- Data is unique in its nature and significance
 - ***Particular*** and ***voluminous***
 - Often asymmetric
 - low value in isolation, high value when aggregated
 - Difficult to protect

Privacy

Security

Misinformation



■ Home ■ News ■ Travel ■ Money ■ Sports ■ Life ■ Tech

Washington/Politics

Inside News

NSA has massive database of Americans' phone calls

Update

The New York Times

© 2017 The New York Times Company

NEW YORK, FRIDAY, JANUARY 20, 2017

\$2.50

WIRETAPPED DATA
USED IN INQUIRY
OF TRUMP AIDES

TRUMP ARRIVES, SET TO ASSUME POWER

In Cabinet Hearings,
Strong Rejection of
Obama's Policies

DOW JONES, A NEWS CORP COMPANY

DJIA ▲ 17873.22 0.25% Nasdaq ▲ 4933.50 0.65% U.S. 10 Yr ▼ 0/32 Yield 1.851% Crude Oil ▲ 49.60 0.55% Euro ▲ 1.1139 0.22%

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New U.S. Study
Fans Cellphone
Cancer Worries



Samsung Adds
More Ads to Its TVs



Verizon Workers
Win Concessions in
Deal to End Strike



103



TECH

LinkedIn 2012 Data Breach May Have Hit Over 100 Million

Professional social network says it will invalidate passwords that weren't changed since

Home / News & Blogs / IT Project Failures

Scathing report slams UK gov't data loss

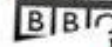
By Michael Krugman | July 1, 2008, 7:33am

Summer
guardian.co.uk

Zurich loses personal details of 51,000 customers

Insurance firm says the data was lost during a routine transfer to South Africa in August last year, but there is no evidence of any misuse

The Press Association
in UK, Thursday 22 October 2008 15:03
high 54. Weather map, Page B4.



Home News Sport Weather TV Radio

ONE-MINUTE WORLD NEWS

Page last updated at 04:12 GMT, Wednesday, 25 June 2008 15:12 UK

E-mail this to a friend

Printable version

Timeline: Child benefits records loss

Two CDs containing personal details of 25m people have been lost by HM Revenue and Customs. Here is how the crisis unfolded.

The Sunday Teleg

Facebook was warned of data risks 7 years ago

By James Titcomb

said the company had no way of know-

Winning feeling



ty Says It Has Thwarted k of Voter Database



Are you ready? Here is all the data Facebook and Google have on you

Dylan Curran

Google knows where you've been

Google knows everything you've ever searched - and deleted

Google has an advertisement profile of you

Google knows all the apps you use

Google has all of your YouTube history

**Facebook stores everything from your s
location**

They can access your webcam and micro

Google knows which events you attend

Google can know your workout routine

And they have years' worth of photos

Google has every email you ever sent

// Manage to gain access to someone's Google account? Perfect, you have a diary of everything that person has done

THE 2022 ONLINE BIG DATA FACTS

5.4b
people online

1.9b
total websites

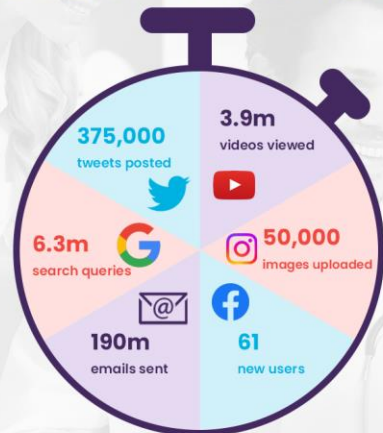
500m
Tweets sent daily

183b
emails sent daily

HOW MUCH DATA IS OUT THERE?

World data is predicted to reach **175ZB** by 2025.
That much data would take one person **1.8 billion years** to download at current internet speeds!

WHAT HAPPENS ONLINE EVERY MINUTE?



Secure your Medical Practice data

 **HealthIT.com.au**

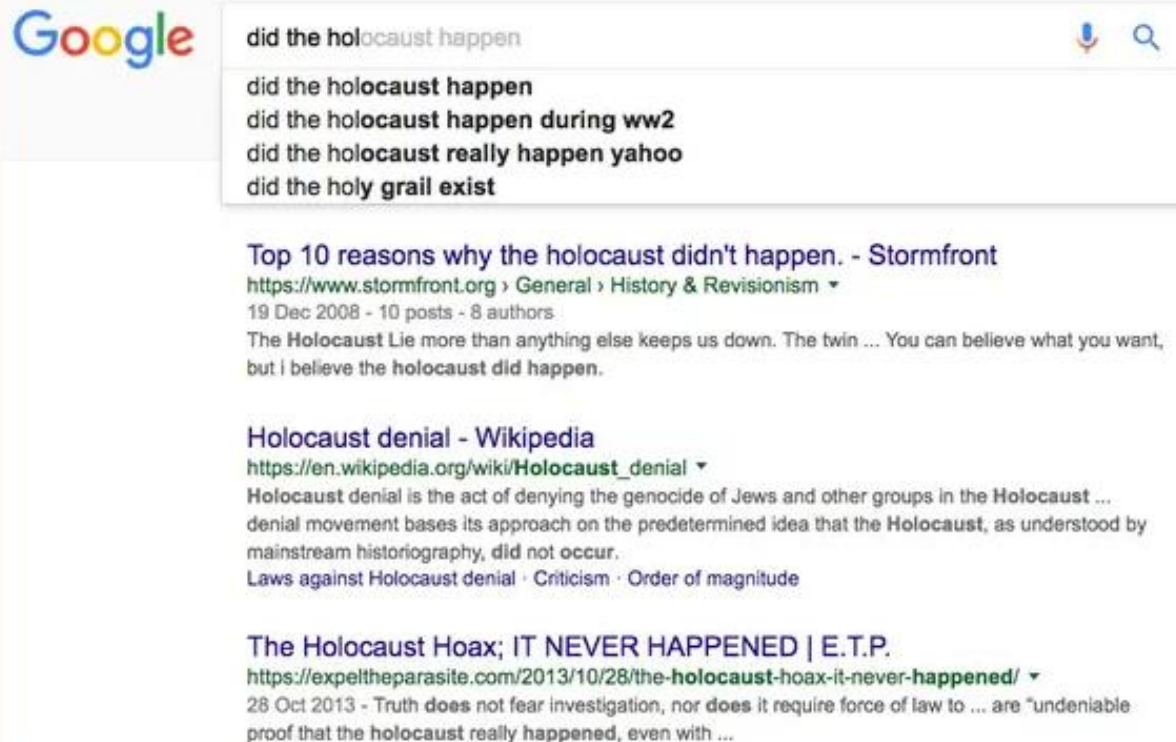
sources: internetlivestats.com, datareportal.com

In 2020, the amount of data on the internet hit 64 zettabytes.

- Let's break down just how big a zettabyte is:
- A gigabyte is 1,024 megabytes
- A terabyte is 1,024 gigabytes
- A petabyte is 1,024 terabytes
- An exabyte is 1,024 petabytes
- A zettabyte is 1,024 exabytes (so that's 1,048,576 petabytes, 1,073,741,824 terabytes, 1,099,511,627,776 gigabytes, or **1,125,899,910,000,000 megabytes!**)

Not all Data is Correct

Last week, we reported how extremist sites 'game' the search engine, boosting their propaganda. In response, the web giant appears to have modified some results, but would like us not to notice



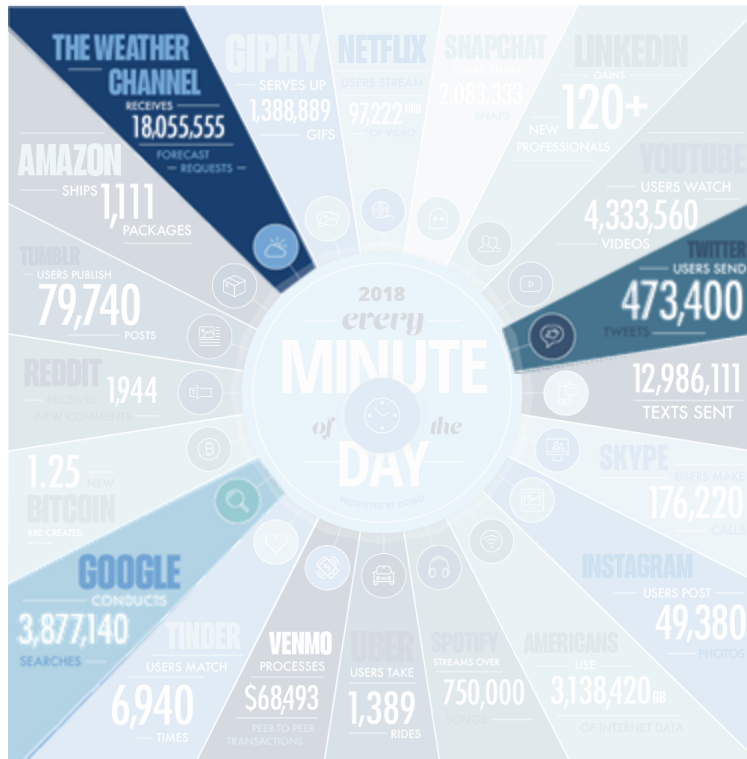
Google is not 'just' a platform. It frames, shapes and distorts how we see the world
Carole Cadwalladr



(From the [Guardian, Dec 2016](#))

Why-R3: The Core of Computing

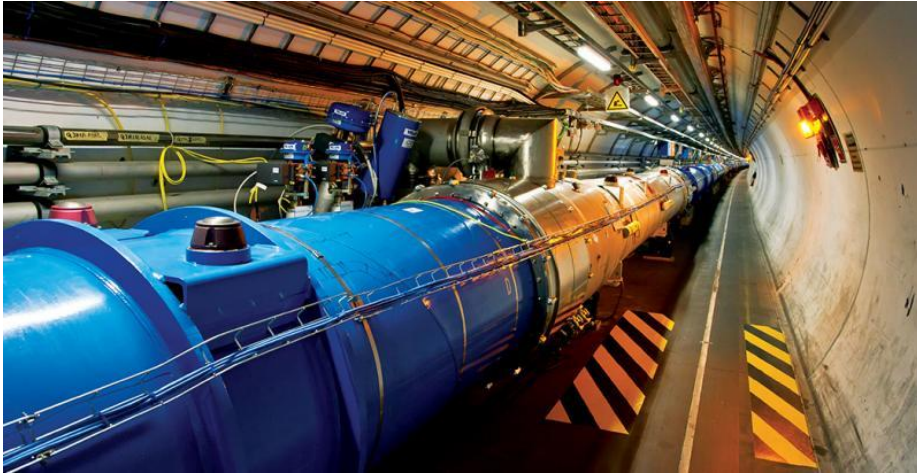
- Data growth will continue to outpace computation
- Systems for Data at Scale: the core of modern computing



Every Minute!

<https://www.domo.com/learn/data-never-sleeps-5>

Scale of Scientific Data



Large Hadron Collider, CERN

- Raw data: 1MB/event. 600,000,000 events/sec.
= 1.9×10^{22} bytes/year = **19 ZettaBytes/year**
- Downsampled: 25GB/sec = 7.88×10^{17} bytes/year = **788 PetaBytes/year**
- Downsampled further: 1050MB/sec = 3.3×10^{16} /year = **33 PetaBytes/year**

<https://home.cern/about/computing/processing-what-record>

Metric prefixes in everyday use			
Text	Symbol	Factor	Power
yotta	Y	1 000 000 000 000 000 000 000 000	10^{24}
zetta	Z	1 000 000 000 000 000 000 000 000	10^{21}
exa	E	1 000 000 000 000 000 000 000	10^{18}
peta	P	1 000 000 000 000 000 000	10^{15}
tera	T	1 000 000 000 000 000	10^{12}
giga	G	1 000 000 000	10^9
mega	M	1 000 000	10^6
kilo	k	1 000	10^3

Forces Driving Data Growth

- Ubiquitous sensors and reporting:
 - Cameras, mobile computing, social media, ...
- Large collaborative science projects
- Philosophy: *More Data* → *More Value*?



Enabling Technology

- **Cheap, Scalable** Data Management Systems



Why-R3: The Core of Computing

- Techniques you learn in this class underlie many topics in computing!

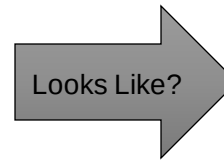


Essential Queries

- **Why** take this class?
- **What** is this class all about?
- **Who** is running this?
- **How** will this class work?

What is this class all about?

- Databases?
 - What is a database?
- Database Management Systems?



Platters on a Disk Drive



Why the Symbol?



Looks
Like?

*“...We must immediately...**attack accounting problems** under the philosophy of handling each business **transaction as it occurs**, rather than under the present condition of **batching techniques**....”*

-- F. J. Wesley IBM Senior Manager



1956: IBM
MODEL 350
RAMAC
First Commercial
Disk Drive
5MB @ 1 ton

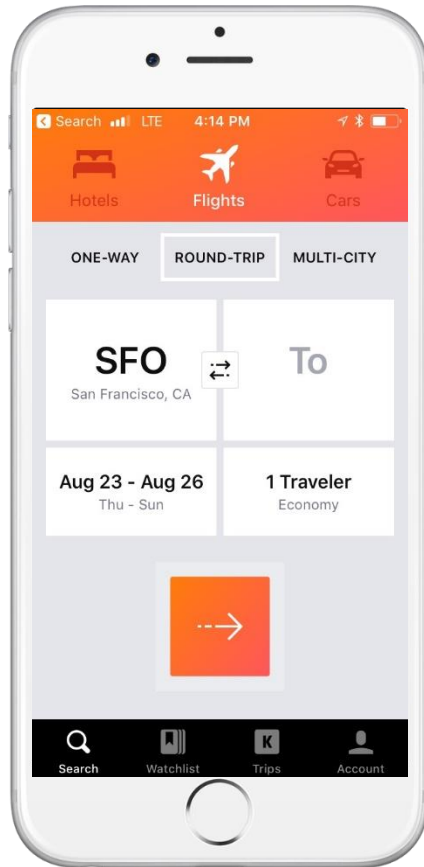
<http://www.computerhistory.org/storageengine/first-commercial-hard-disk-drive-shipped>

Is This a Database?

- Rolodex (旋转式名片卡片存储设备)
- Alphabetically ordered cards
- Indexed access by first letter



Is This a Database?



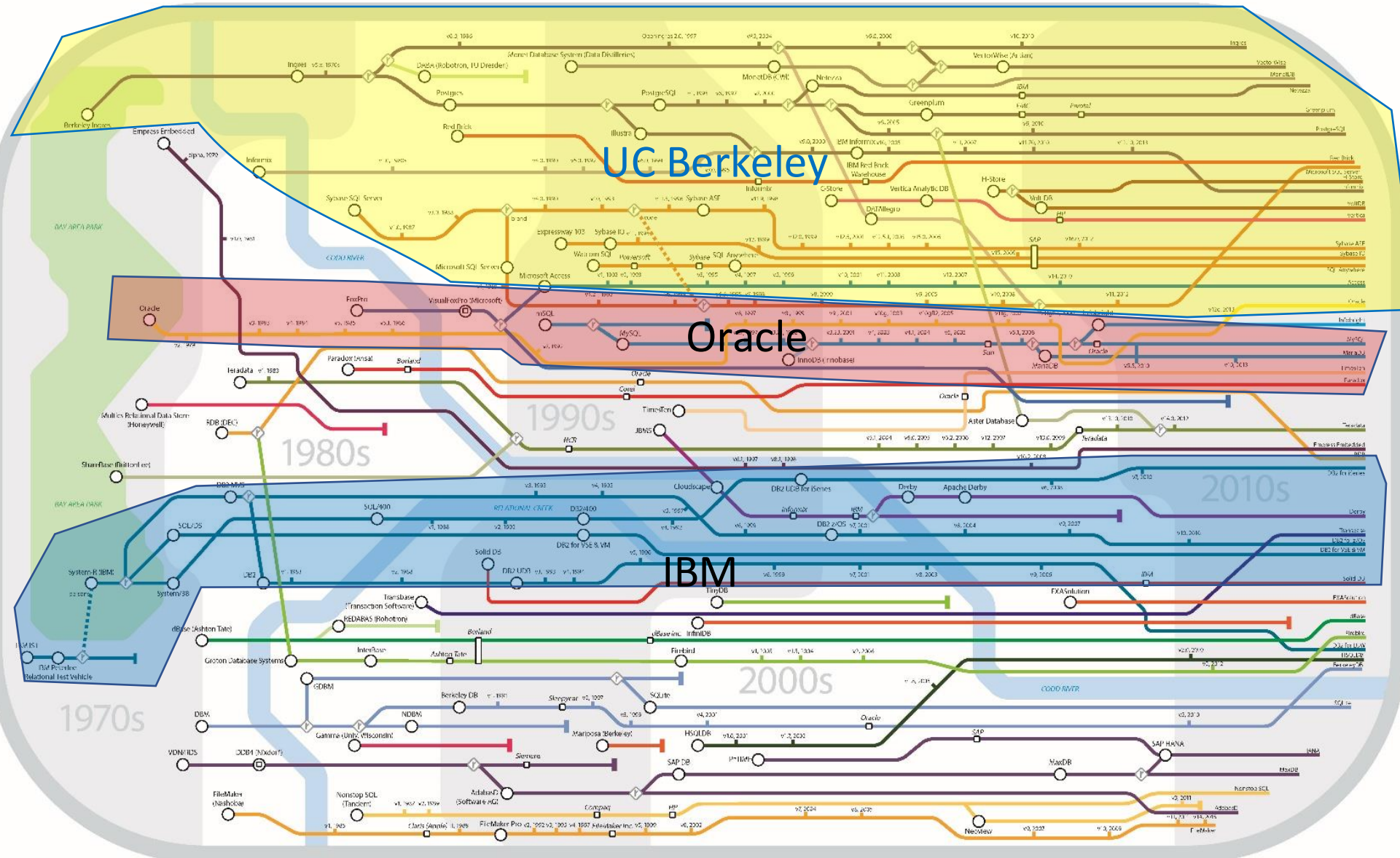
- Airline reservation systems were one of the earliest pervasive consumer uses of database systems.
 - IBM/American Airlines' SABRE system, 1964.
 - “Semi-Automated Business Research Environment”
 - Travelocity.com a direct descendant of SABRE
 - Acquired by Expedia, 1/2015



What is a Database?

- A database is a large, organized collection of data.
- Sometimes confused with a Database Management System (DBMS)
 - *A DBMS is software that **stores, manages, and facilitates access to data.***

Genealogy of Relational Database Management Systems



Key to lines and symbols

- Publishing Date
- Acquisition
- Versions
- Discontinued
- ◇ Branch intellectual and/or code
- Crossing lines have no special semantics

Felix Naumann, Jane Bauckmann, Cauda Eiser, Jan-Peer Rudolph, Fabian Tschirschnitz
 Contact: Hasso Plattner Institut, Potsdam, Felix.Naumann@hpi.uni-potsdam.de
 Design: Alexander Sandt, Grafik-Design, Hamburg
 Version 5.0, October 2015
http://www.hpi.uni-potsdam.de/naumann/projekte/rdbms_genealogy.html

Relational DBMSs

- Traditionally DBMS referred to relational databases



- **RDBMS** is a more appropriate term
- **SQL** data description and manipulation language
- **ACID** transaction consistency
- **Durable** writes (prevent data loss)
- **Mature** technologies ...



Ranking of DBMS Technologies 2023 vs 2024

Rank			DBMS	Database Model	Score		
Sep 2023	Aug 2023	Sep 2022			Sep 2023	Aug 2023	Sep 2022
1.	1.	1.	Oracle +	Relational, Multi-model ⓘ	1240.88	-1.22	+2.62
2.	2.	2.	MySQL +	Relational, Multi-model ⓘ	1111.49	-18.97	-100.98
3.	3.	3.	Microsoft SQL Server +	Relational, Multi-model ⓘ	902.22	-18.60	-24.08
4.	4.	4.	PostgreSQL +	Relational, Multi-model ⓘ	620.75	+0.37	+0.29
5.	5.	5.	MongoDB +	Document, Multi-model ⓘ	439.42	+4.93	-50.21
6.	6.	6.	Redis +	Key-value, Multi-model ⓘ	163.68	+0.72	-17.79
7.	7.	7.	Elasticsearch				
8.	8.	8.	IBM Db2				
9.	↑ 10.	↑ 10.	SQLite +				
10.	↓ 9.	↓ 9.	Microsoft Ac				

423 systems in ranking, September 2024

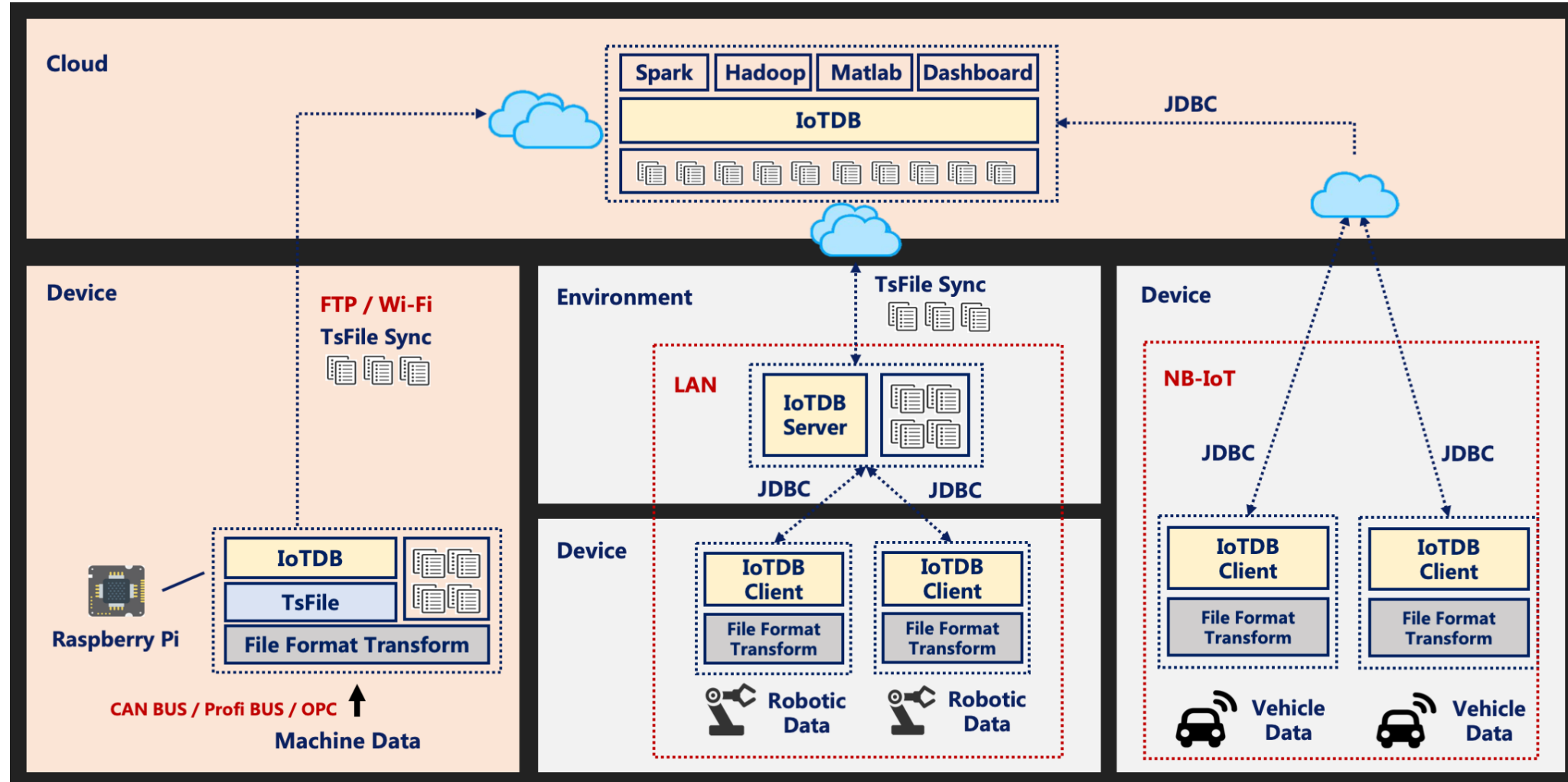
Rank			DBMS	Database Model	Score		
Sep 2024	Aug 2024	Sep 2023			Sep 2024	Aug 2024	Sep 2023
1.	1.	1.	Oracle +	Relational, Multi-model ⓘ	1286.59	+28.11	+45.72
2.	2.	2.	MySQL +	Relational, Multi-model ⓘ	1029.49	+2.63	-82.00
3.	3.	3.	Microsoft SQL Server +	Relational, Multi-model ⓘ	807.76	-7.41	-94.45
4.	4.	4.	PostgreSQL +	Relational, Multi-model ⓘ	644.36	+6.97	+23.61
5.	5.	5.	MongoDB +	Document, Multi-model ⓘ	410.24	-10.74	-29.18
6.	6.	6.	Redis +	Key-value, Multi-model ⓘ	149.43	-3.28	-14.26
7.	7.	↑ 11.	Snowflake +	Relational	133.72	-2.25	+12.83
8.	8.	↓ 7.	Elasticsearch	Search engine, Multi-model ⓘ	128.79	-1.04	-10.20
9.	9.	↓ 8.	IBM Db2	Relational, Multi-model ⓘ	123.05	+0.04	-13.67
10.	10.	↓ 9.	SQLite +	Relational	103.35	-1.44	-25.85



Change = Opportunity !

- The DBMS world is rapidly changing
 - Will discuss these changes towards end of the course
 - Our textbook is rather out of date
- Opportunity!
 - You can shape the future of DBMSs
- We won't follow the textbook slavishly.

Example: IoTDB-THU





Focus & Goal

- Focus: **Foundational System Principles**
 - Basic ideas and components
 - How to compose those components into a technology stack
- Goal:
 - You will be able to **use** existing & **build new** DBMS technologies!



Databases



Main Contents

In this course, we will learn the **basic concepts**, **principles** and **applications** of database systems, especially the *relational* database systems. The contents mainly include :

- The data models (Ch2)
- SQL language and user interfaces (Ch3)
- Key principles of DBMS (mainly architecture, query optimization, concurrency control, recovery, etc.) (Ch4)
- The security and integrity constraints of database (Ch5)
- Database Design (Ch6)
- New Research and Application Fields



Table of Contents

1. Introduction

The history, classification, and main research contents of database systems;
The database system; the concepts of data model

2. Data Model*

Hierarchical and network model; Relational model; ER model; Object-Oriented model and other data models

3. User Interfaces and SQL Language

User interface; SQL language, including QL, DDL, DCL, DML, view, embedded SQL and dynamic SQL, etc.



Table of Contents

4. Database Management Systems

The architecture of database systems, query optimization, file structure and index, transaction management, concurrency control, recovery mechanism

5. The Security and Integrity Constrain

The security model of database system; Integrity constrain and its expression, implementing method, assertion, trigger

6. Database Design*

Design procedure; ER graph; Normalization of Relational Schema



1. Introduction

What Is Database?

**What Is Database management
System (DBMS)?**



Database & DBMS

- A very large, integrated collection of data.
- Models real-world *enterprise*.
 - Entities (e.g., students, courses)
 - Relationships (e.g., electives)
- A *Database Management System (DBMS)* is a software package designed to store and manage databases.



Why Use a DBMS?



Why Use a DBMS?

- *Data independence* and efficient access.
- Reduced application development time.
- Data *integrity and security*.
- Uniform data administration.
- *Concurrent access, recovery from crashes*.

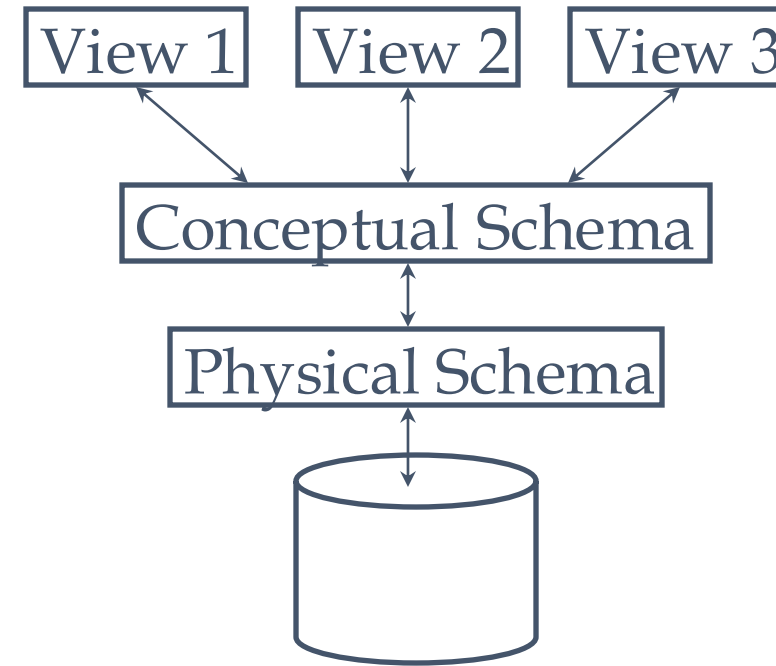


Data, Data Model and Data Schema

- **Data** are symbols for describing the things of real world. They are existing form of information.
- A **data model** is a collection of concepts and definitions for describing data.
- A **schema** is a description of a particular collection of data, using a given data model.
- The **relational model of data** is the most widely used model today.
 - Main concept: **relation**, basically a table with rows and columns.
 - Every relation has a **schema**, which describes the columns, or fields.

Levels of Abstraction: ANSI-SPARC Architecture

- ⑩ Many **views**, single **conceptual (logical) schema** and **physical schema**.
 - ⑩ Views describe how users see the data.
 - ⑩ Conceptual schema defines logical structure
 - ⑩ Physical schema describes the files and indexes used.



* *Schemas are defined using **DDL**; data is modified/queried using **DML**.*



Data Definition Language (DDL)

- Specification notation for defining the database schema

Example: **create table** *instructor* (
 ID **char**(5),
 name **varchar**(20),
 dept_name **varchar**(20),
 salary **numeric**(8,2))

- DDL compiler generates a set of table templates stored in a **data dictionary**
- Data dictionary contains **metadata** (i.e., data about data)
 - Database schema
 - Integrity constraints
 - Primary key (ID uniquely identifies instructors)
 - Authorization
 - Who can access what



Data Manipulation Language (DML)

- Language for accessing and updating the data organized by the appropriate data model
 - DML also known as *query language*
- There are basically two types of data-manipulation language
 - *Procedural DML* -- require a user to specify what data are needed and how to get those data.
 - *Declarative DML*-- require a user to specify what data are needed without specifying how to get those data.
- Declarative DMLs are usually easier to learn and use than are procedural DMLs.
- Declarative DMLs are also referred to as non-procedural DMLs
- The portion of a DML that involves information retrieval is called a *query* language.



SQL Structured Query Language

- SQL query language is nonprocedural. A query takes as input several tables (possibly only one) and always returns a single table.
- Example to find all instructors in Comp. Sci. dept

```
select name  
from instructor  
where dept_name = 'Comp. Sci.'
```

- SQL is **NOT** a Turing machine equivalent language
 - To be able to compute complex functions SQL is usually embedded in some higher-level language
- Application programs generally access databases through one of
 - Language extensions to allow **embedded SQL**
 - Application **program interface** (e.g., ODBC/JDBC) which allow SQL queries to be sent to a database



Database Access from Application Program

- Non-procedural query languages such as SQL are not as powerful as a universal Turing machine.
- SQL does not support actions such as input from users, output to displays, or communication over the network.
- Such computations and actions must be written in a *host language*, such as C/C++, Java or Python, with embedded SQL queries that access the data in the database.
- *Application programs* -- are programs that are used to interact with the database in this fashion.





History of Database Systems

- 1950s and early 1960s:
 - Data processing using magnetic tapes for storage
 - Tapes provided only sequential access
 - Punched cards for input
- Late 1960s and 1970s:
 - Hard disks allowed direct access to data
 - ***Network and hierarchical data models in widespread use***
 - ***Ted Codd defines the relational data model***
 - Would win the ACM Turing Award for this work
 - IBM Research begins System R prototype
 - UC Berkeley (Michael Stonebraker) begins Ingres prototype
 - Oracle releases first commercial relational database
 - High-performance (for the era) transaction processing



History of Database Systems (Cont.)

- 1980s:
 - *Research relational prototypes evolve into commercial systems*
 - *SQL becomes industrial standard*
 - Parallel and distributed database systems
 - Wisconsin, IBM, Teradata
 - *Object-oriented database systems*
- 1990s:
 - Large decision support and data-mining applications
 - Large multi-terabyte data warehouses
 - Emergence of Web commerce



History of Database Systems (Cont.)

- 2000s
 - Big data storage systems
 - Google BigTable, Yahoo PNuts, Amazon,
 - *“NoSQL” systems.*
 - Big data analysis: beyond SQL
 - Map reduce and friends
- 2010s
 - SQL reloaded
 - SQL front end to Map Reduce systems
 - Massively parallel database systems
 - Multi-core main-memory databases



Classification

(1) According to the development of data model

- No management (before 1960'): Scientific computing
- File system: Simple data management
- Demand of data management growing continuously, DBMS emerged.
 - 1964, the first DBMS (American): IDS, network
 - 1969, the first commercial DBMS of IBM, hierarchical
 - 1970, E.F.Codd(IBM) bring forward relational data model
 - Other data model: Object Oriented, deductive, ER, ...



Classification

2) According to the development of DBMS architectures

- Centralized database systems
- Parallel database systems
- Distributed database systems
- Mobile database systems

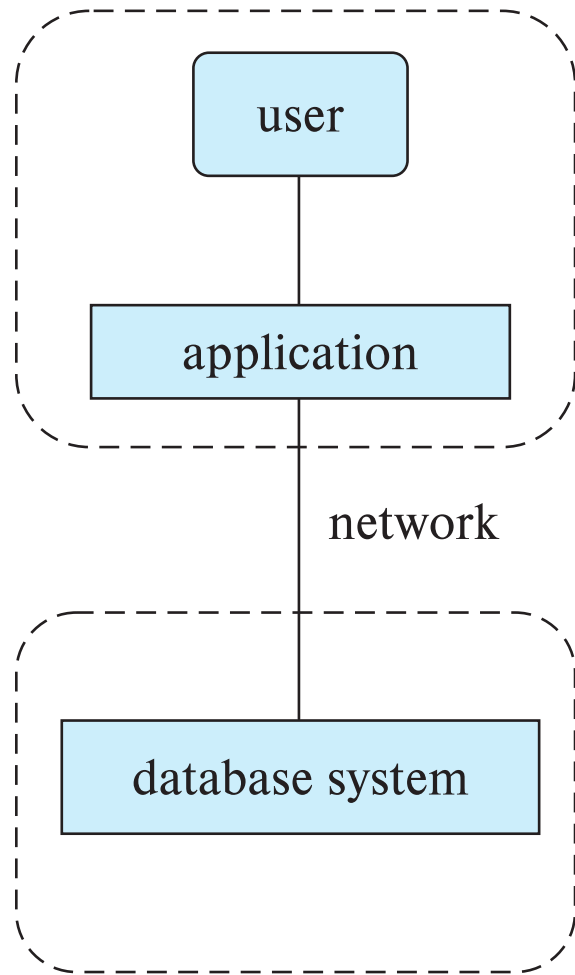


Classification

(3) According to the development of architectures of application systems based on databases

- Centralized structure : Host + Terminal
- Distributed structure
- Client/Server structure
- Three tier/multi-tier structure
- Mobile computing
- Grid computing / Cloud computing

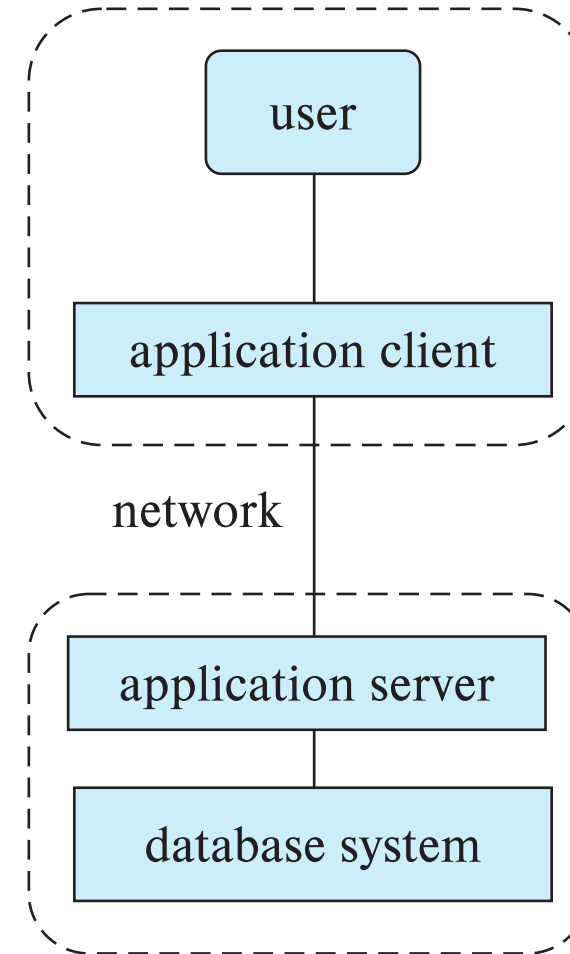
Two-tier and three-tier architectures



(a) Two-tier architecture

client

server



(b) Three-tier architecture

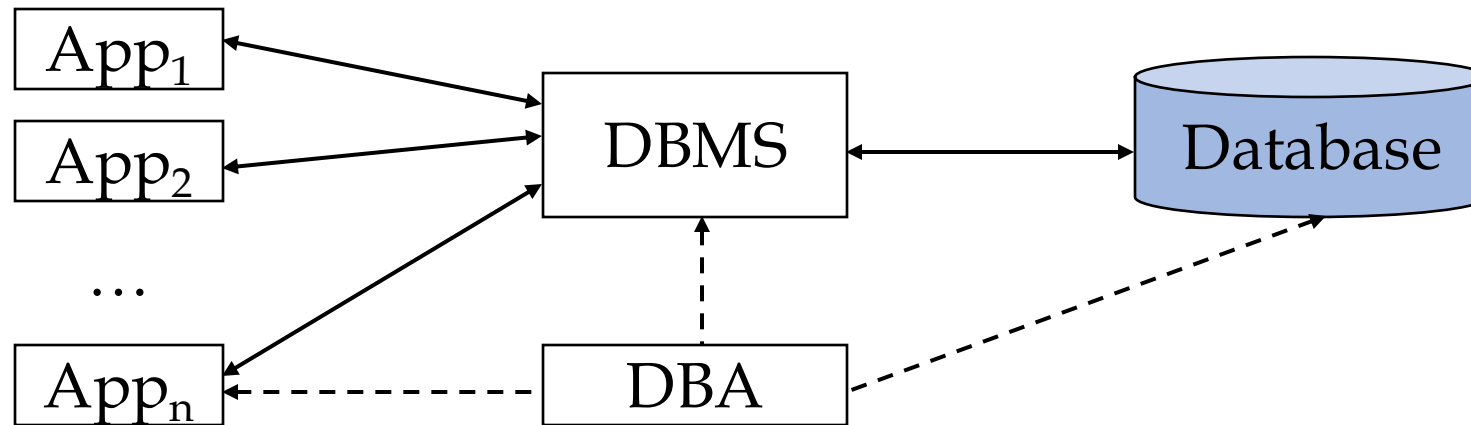


Classification

(4) According to the expanding of application fields

- On-line Transaction Processing (OLTP)
- Engineering Database
- Multimedia Database
- Temporal Database
- Spatial Database
- Data Warehouse, On-line Analytical Processing (OLAP), Data Mining
- Deductive Database, Knowledge Management
-

Database System

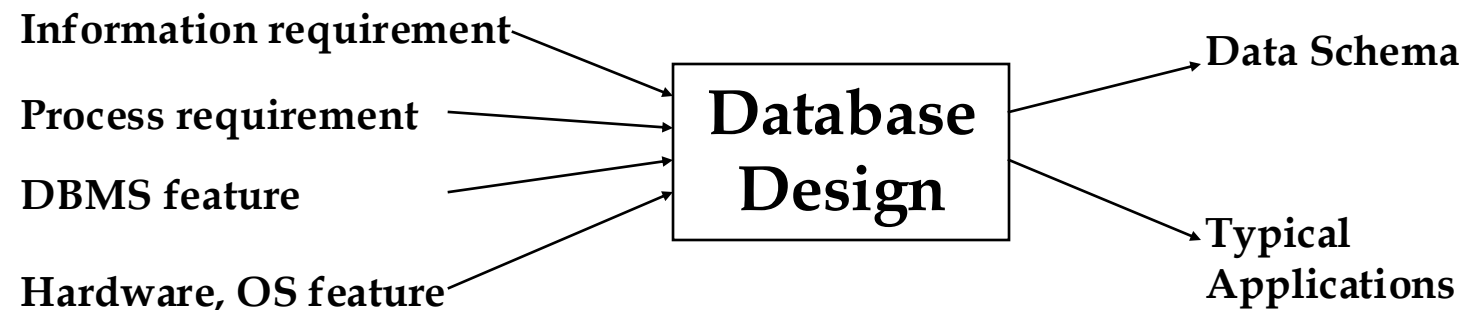


- **Applications + DBMS + Database + DBA**
- **DBMS is the core of database system**
 - High level user interfaces
 - Query processing and optimization
 - Catalog management
 - Concurrency control and Recovery
 - Integrity constraints checking
 - Access control



Life cycle of database systems

- Database system planning
- Database designing
- Database establishing and loading
- Database running, managing and maintaining
- Database extending and restructuring





Summary

- What Is Database, DBMS, Database System, and relational DBMS?
- Why DBMS?
- What's Data, Data Model and Data Schema?
- What's ANSI-SPARC Architecture and how it supports data independence?